

Cryptography Toolkit — Quick Reference

Everything defined in `crypto.tex`, shown in use.

1 Theorem environments

All theorem-like environments share one counter numbered within the section, so cross-references stay unambiguous.

Definition 1.1 (Perfect secrecy). An encryption scheme $(\text{Gen}, \text{Enc}, \text{Dec})$ over message space $\mathcal{M}$ is **perfectly secret** if for every $m_0, m_1 \in \mathcal{M}$ and every ciphertext $c \in \mathcal{C}$,

$$\Pr[\text{Enc}_k(m_0) = c] = \Pr[\text{Enc}_k(m_1) = c], \quad (1)$$

where $k \xleftarrow{\$} \mathcal{K}$.

Theorem 1.2 (Shannon). *If a scheme over $\mathcal{M}$ with key space $\mathcal{K}$ is perfectly secret, then $|\mathcal{K}| \geq |\mathcal{M}|$.*

Proof. Suppose $|\mathcal{K}| < |\mathcal{M}|$. Fix a ciphertext c with $\Pr[C = c] > 0$ and let $\mathcal{M}(c) = \{\text{Dec}_k(c) : k \in \mathcal{K}\}$. Then $|\mathcal{M}(c)| \leq |\mathcal{K}| < |\mathcal{M}|$, so some $m' \in \mathcal{M}$ lies outside $\mathcal{M}(c)$ and $\Pr[\text{Enc}_k(m') = c] = 0$, contradicting Definition 1.1. $\square$

Construction 1.3 (One-time pad). Over $\mathcal{M} = \{0, 1\}^n$: **Gen** outputs $k \xleftarrow{\$} \{0, 1\}^n$; $\text{Enc}_k(m) = k \oplus m$; $\text{Dec}_k(c) = k \oplus c$.

Lemma 1.4. *Construction 1.3 satisfies Definition 1.1 and meets the bound of Theorem 1.2 with equality.*

Remark 1.5. Available: `definition`, `theorem`, `lemma`, `corollary`, `proposition`, `claim`, `construction`, `example`, `remark`, `proof`.

2 Algorithms

Algorithm 1 Square-and-multiply modular exponentiation.

Input: $a \in \mathbb{Z}_n$, exponent $e = \sum_{i=0}^t e_i 2^i$, modulus n

Output: $a^e \bmod n$

```

1:  $r \leftarrow 1$ 
2:  $b \leftarrow a \bmod n$ 
3: for  $i \leftarrow 0$  to  $t$  do
4:   if  $e_i = 1$  then
5:      $r \leftarrow r \cdot b \bmod n$ 
6:   end if
7:    $b \leftarrow b^2 \bmod n$ 
8: end for
9: return  $r$ 

```

Algorithm 1 runs in $O(\log e)$ modular multiplications.

Algorithm 2 Extended Euclidean algorithm.

Input: $a, b \in \mathbb{N}$ with $a \geq b > 0$

Output: (d, x, y) such that $d = \gcd(a, b) = ax + by$

```

1: function EXTgcd( $a, b$ )
2:   if  $b = 0$  then
3:     return  $(a, 1, 0)$ 
4:   end if
5:    $(d, x', y') \leftarrow \text{EXTgcd}(b, a \bmod b)$ 
6:   return  $(d, y', x' - \lfloor a/b \rfloor \cdot y')$ 
7: end function

```

3 Protocol diagrams

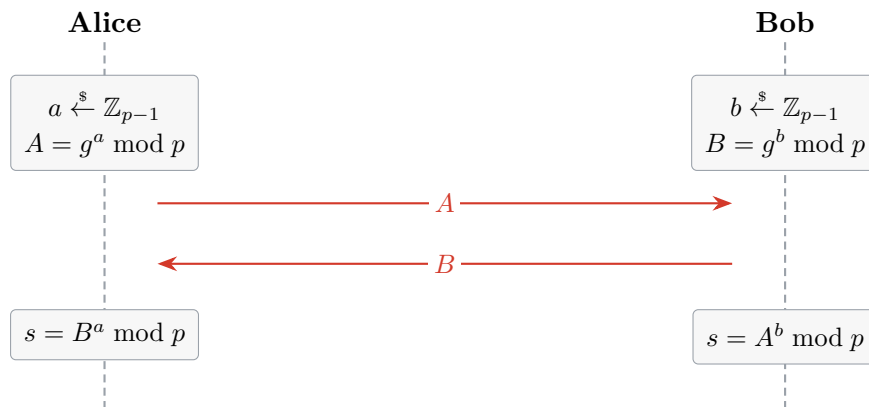


Figure 1. Diffie–Hellman key exchange over $\mathbb{Z}_p^*$.

Both parties in figure 1 derive $s = g^{ab} \bmod p$. Styles available inside a `protocol` environment: `party`, `lifeline`, `msg`, `msglabel`, `compute`.

4 Notation macros

Table 1. Notation macros defined in `crypto.tex`.

Macro	Output	Use
<code>\Zn{n}</code>	$\mathbb{Z}_n$	Integers modulo n
<code>\Zns{p}</code>	$\mathbb{Z}_p^*$	Multiplicative group
<code>\Fq{q}</code>	$\mathbb{F}_q$	Finite field
<code>\bitstr{n}</code>	$\{0, 1\}^n$	Bit strings of length n
<code>\bits</code>	$\{0, 1\}$	The set $\{0, 1\}$
<code>\xor</code>	$a \oplus b$	Bitwise XOR
<code>\concat</code>	$a \parallel b$	Concatenation
<code>\getsr</code>	$x \xleftarrow{\$} S$	Uniform sampling
<code>\Enc \Dec</code>	Enc, Dec	Encryption / decryption
<code>\Gen</code>	Gen	Key generation
<code>\Sign \Vrfy</code>	Sign, Vrfy	Signing / verification
<code>\Mac</code>	Mac	Message authentication
<code>\Hash</code>	H	Hash function
<code>\adv</code>	$\mathcal{A}$	Adversary
<code>\secpa</code>	λ	Security parameter
<code>\negl</code>	$\text{negl}(\lambda)$	Negligible function
<code>\Adv</code>	Adv	Advantage
<code>\ord</code>	$\text{ord}(g)$	Order of an element
<code>\lcm</code>	$\text{lcm}(a, b)$	Least common multiple
<code>\Prob{...}</code>	$\Pr[X = x]$	Probability
<code>\abs{...}</code>	$ \mathcal{K} $	Absolute value / cardinality
<code>\ceil \floor</code>	$\lceil x \rceil, \lfloor x \rfloor$	Rounding
<code>\msgspace</code>	$\mathcal{M}$	Message space
<code>\keyspace</code>	$\mathcal{K}$	Key space
<code>\ctspace</code>	$\mathcal{C}$	Ciphertext space

A worked line combining several of them:

$$\text{Adv}_{\Pi}^{\text{ind}}(\mathcal{A}) = |\Pr[\mathcal{A}(\text{Enc}_k(m_0)) = 1] - \Pr[\mathcal{A}(\text{Enc}_k(m_1)) = 1]| \leq \text{negl}(\lambda). \quad (2)$$